DriftGuard Clinical | Movement Risk Screening Report

AI-Assisted Biomechanical Assessment & Baseline Comparison Platform

NON-DIAGNOSTIC SCREENING AID: *This report presents computer-vision movement screening proxies and baseline comparisons. It is intended solely for clinical decision support by licensed doctors and physiotherapists, and does NOT constitute a formal medical diagnosis or ACL injury confirmation.*

MEASUREMENT PROVENANCE & HARDWARE DISCLOSURE: *Camera Setup: Monocular RGB (MediaPipe 3D) | IMU Status: IMU integration available — hardware not connected | Kinematics: Pose-Derived Proxies | Mocap Validation: Not Validated against Mocap Reference*

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1. Athlete & Session Metadata

Athlete Name: Demo Athlete

Session ID: ses_f4b5911b19fa

Athlete ID: ath_6f7a056c760f

Assessment Date: 2026-10-09

Sport Profile: Basketball

Camera Setup: frontal+sagittal

Dominant Leg: Unknown

Overall Risk Band: **LOW**

2. Observed Kinematic Summary (Mean / Event Peak)

Biomechanical Metric	Left Limb	Right Limb	Limb Symmetry (LSI)
Knee FPPA Valgus Proxy	-19.6°	7.8°	200.0%
Sagittal Knee Flexion	19.6°	7.8°	86.0%
Trunk Lateral Lean	52.0°	N/A	N/A

3. Confirmed Risk Flags & Literature Evidence

Feature	Risk Band	Deviation	Clinical Detail / Citation
knee_flexion_left	**Moderate**	+-1.8 IQR	**Elevated movement-risk pattern: Left knee landing flexion has been moderately elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in left knee landing flexion, consistently below their established baseline median of 41.0 degrees (+/-1.8 degrees) by 1.8x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Shin et al. 2009 (reduced landing knee flexion as ACL-injury risk factor)

knee_flexion_left	**Moderate**	+-1.8 IQR	**Elevated movement-risk pattern: Left knee landing flexion has been moderately elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in left knee landing flexion, consistently below their established baseline median of 41.0 degrees (+/-1.8 degrees) by 1.8x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Shin et al. 2009 (reduced landing knee flexion as ACL-injury risk factor)
fppa_right	**High**	+-3.5 IQR	**Elevated movement-risk pattern: Right knee valgus angle (fppa) has been substantially elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in right knee valgus angle (FPPA), consistently below their established baseline median of 7.5 degrees (+/-1.6 degrees) by 3.5x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Hewett et al. 2005; Bates et al. 2020 (knee abduction / valgus angle as ACL-injury risk factor)
knee_flexion_left	**Moderate**	+-7.7 IQR	**Elevated movement-risk pattern: Left knee landing flexion has been moderately elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in left knee landing flexion, consistently below their established baseline median of 40.4 degrees (+/-1.8 degrees) by 7.7x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Shin et al. 2009 (reduced landing knee flexion as ACL-injury risk factor)
fppa_right	**High**	+-3.6 IQR	**Elevated movement-risk pattern: Right knee valgus angle (fppa) has been substantially elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in right knee valgus angle (FPPA), consistently below their established baseline median of 7.7 degrees (+/-1.6 degrees) by 3.6x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Hewett et al. 2005; Bates et al. 2020 (knee abduction / valgus angle as ACL-injury risk factor)
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fppa_right	**High**	+-2.9 IQR	**Elevated movement-risk pattern: Right knee valgus angle (fppa) has been substantially elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in right knee valgus angle (FPPA), consistently below their established baseline median of 7.4 degrees (+/-1.9 degrees) by 2.9x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Hewett et al. 2005; Bates et al. 2020 (knee abduction / valgus angle as ACL-injury risk factor)
knee_flexion_left	**Moderate**	+-1.0 IQR	**Elevated movement-risk pattern: Left knee landing flexion has been moderately elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in left knee landing flexion, consistently below their established baseline median of 40.4 degrees (+/-13.3 degrees) by 1.0x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Shin et al. 2009 (reduced landing knee flexion as ACL-injury risk factor)
fppa_right	**Moderate**	+-2.9 IQR	**Elevated movement-risk pattern: Right knee valgus angle (fppa) has been moderately elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in right knee valgus angle (FPPA), consistently below their established baseline median of 7.4 degrees (+/-1.9 degrees) by 2.9x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Hewett et al. 2005; Bates et al. 2020 (knee abduction / valgus angle as ACL-injury risk factor)
fppa_right	**Moderate**	+-2.5 IQR	**Elevated movement-risk pattern: Right knee valgus angle (fppa) has been moderately elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in right knee valgus angle (FPPA), consistently below their established baseline median of 5.7 degrees (+/-1.9 degrees) by 2.5x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Hewett et al. 2005; Bates et al. 2020 (knee abduction / valgus angle as ACL-injury risk factor)
fppa_right	**Moderate**	+-2.6 IQR	**Elevated movement-risk pattern: Right knee valgus angle (fppa) has been moderately elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in right knee valgus angle (FPPA), consistently below their established baseline median of 6.0 degrees (+/-1.9 degrees) by 2.6x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Hewett et al. 2005; Bates et al. 2020 (knee abduction / valgus angle as ACL-injury risk factor)

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4. Clinician Observations & Neuromuscular Retraining Notes

No clinician notes recorded for this session yet.

Assessing Clinician Signature: _____ **Date:** _____